EEFE 530 - Problem Set 4

Difference-in-Differences Simulation

EEFE 530: Applied Microeconometrics

2026-03-15

Overview

This problem set examines new methods using robust to treatment heterogeneity in settings with staggered adoption. This will be a group exercise where each group is assigned one paper to cover. You will present this in-class on Monday March 15. Your presentation should be no more than 20 minutes.

Group assignments:

1. Difference-in-Differences with multiple time periods & Estimating dynamic treatment effects in event studies with heterogeneous treatment effects
   - Jieun Hyun, Yuxiao Zhu & Yu Ping Chang (this group has three people and will cover two papers that are pretty similar)
2. Revisiting Event-Study Designs: Robust and Efficient Estimation
   - Jiahui Guo & Chia-Chen Hsu
3. Efficient Estimation for Staggered Rollout Designs
   - Junxian Tang & Xiaofei Wang
4. Two-Stage Differences in Differences
   - Chunyi Zhang & Samyar Darzilarijani
5. A Local Projections Approach to Difference-in-Differences
   - Jiahi Chen & Chung-Kang Lo

The links above reference the papers. All the papers have software packages made by the authors or other academics to implement the estimators.

In the ps4 repository on the public course page there are six simulated datasets:

- `data/data_A.RData`
- `data/data_B.RData`
- `data/data_C.RData`
- `data/data_D.RData`
- `data/data_E.RData`
- `data/data_F.RData`

Submission

You should submit your presentation before class on canvas or GitHub Classroom. Each group member should submit a copy of the presentation.

Push to GitHub Classroom:

- `ps4_<LAST_NAME>.qmd`
- rendered output: `submission_<LAST_NAME>.html` (or `submission_<LAST_NAME>.pdf`)
- any helper files you used (figures, scripts)

Canvas: submit a pdf on Canvas

Write your code inline as shown below

```r
# Example code
treat.test <- c(1,2,3)
control.test <- c(0,1,2)

att.test <- mean(treat.test) - mean(control.test)
```

1 Explain and present the new method

Generate slides and present to the class the following topics.

1. What is the estimator?
2. How does it fix the problem of staggered adoption?
3. What are the assumptions necessary for causal identification?
4. What are pros and cons of the estimator?

2 Estimation on the simulated data

For each simulated dataset conduct the following analysis:

1. Estimate the ATT using TWFE and the new estimator
2. Estimate and plot an event study model using TWFE and the new estimator
3. Describe the differences between the two estimators and evaluate whether the assumptions hold